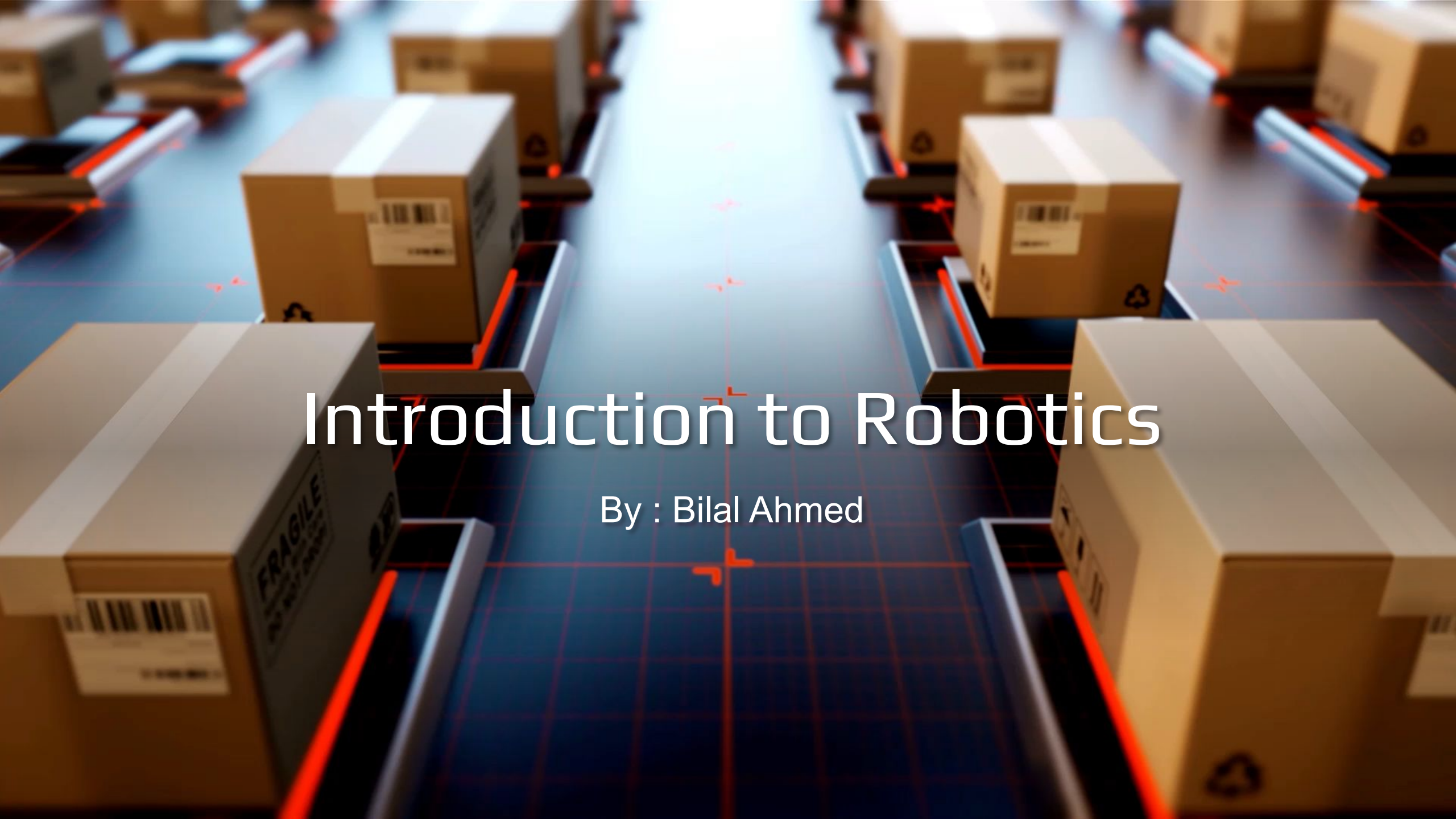
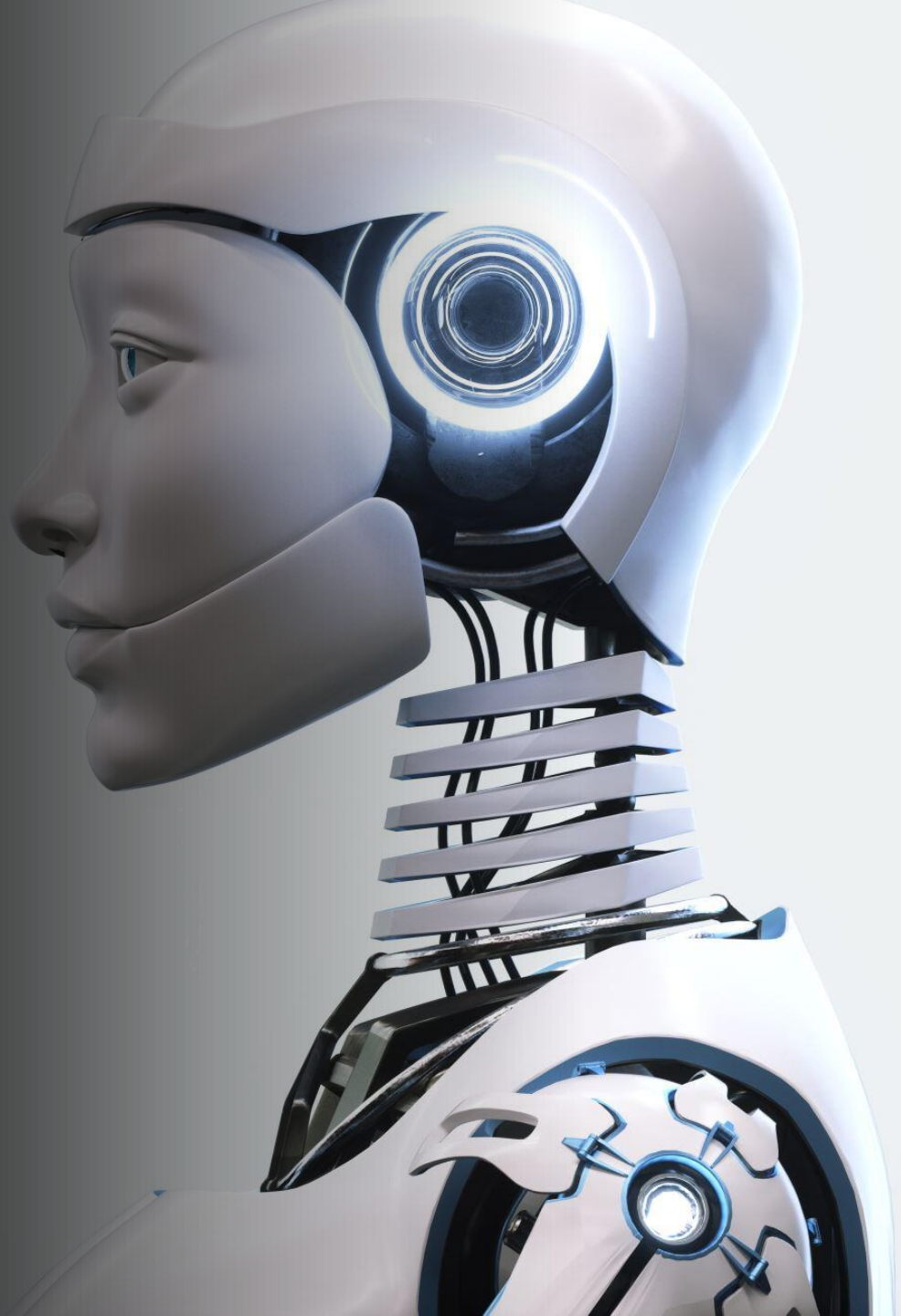


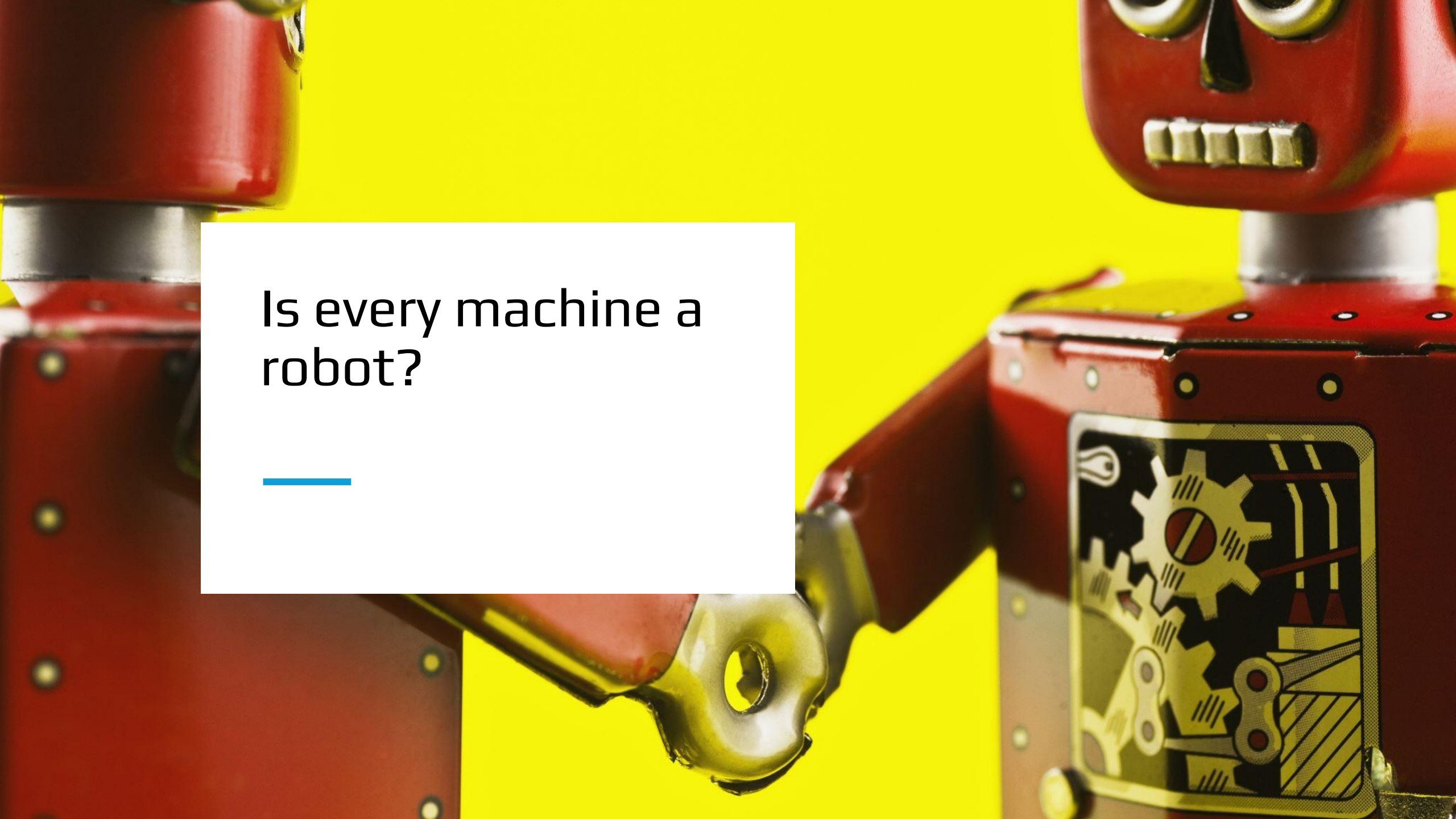
Introduction to Robotics

By : Bilal Ahmed

What is a robot?

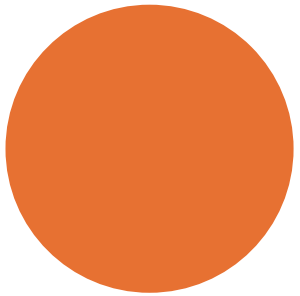
- A robot is an intelligent machine that can sense, think, and act in the physical world
- A robot is basically a combination of:
  - sensors (eyes/ears)
  - controller (brain)
  - motors (muscles)



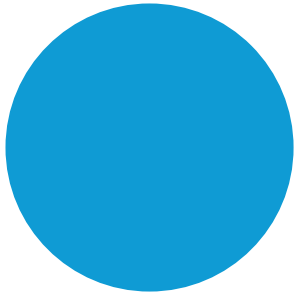


Is every machine a robot?

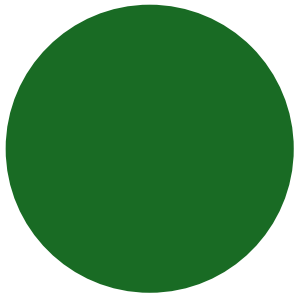
Consider following



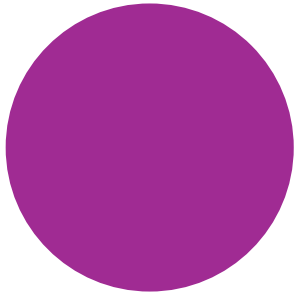
Microwave oven that runs on fixed time and temperature settings.



A 2-wheel robot with ultrasonic sensor.



A robotic vacuum cleaner that detects obstacles and changes movement path dynamically.



Washing machine with preset cycles (wash, rinse, spin).

Robotics vs Automation

Automation focuses on *doing tasks automatically under fixed rules.*

Robotics focuses on *machines that can sense, decide, and act in the real world.*

Basic Idea: Automation follows instructions. Robots can also understand the environment.

Comparison

Feature	Automation	Robotics
Definition	System that performs tasks automatically based on predefined rules	Intelligent machine that can sense, decide, and act in the physical world
Environment Awareness	Limited or none	High (uses sensors)
Decision Making	Rule-based (pre-programmed)	AI-based or adaptive decision-making
Flexibility	Low (fixed tasks)	High (can adapt to changes)
Physical Interaction	May or may not interact with physical world	Always interacts with physical world
Sensors	Rare or minimal	Essential component
Examples	Washing machine, elevator system, factory conveyor belt	Self-driving car, drone, humanoid robot
Complexity	Generally simpler	More complex (hardware + software + AI)
Learning Ability	No learning	Can include learning (AI/ML systems)

Examples

Automatic BUT NOT Robot

- Elevator system (moves automatically between floors)
- Microwave oven with preset programs
- Traffic light system
- Washing machine

Automatic AND Robot

- Autonomous vacuum cleaner (e.g., Roomba)
- Self-driving car
- Obstacle-avoiding robot
- Industrial robotic arm with vision system

Can we upgrade a washing machine to a robot?



The Upgrade

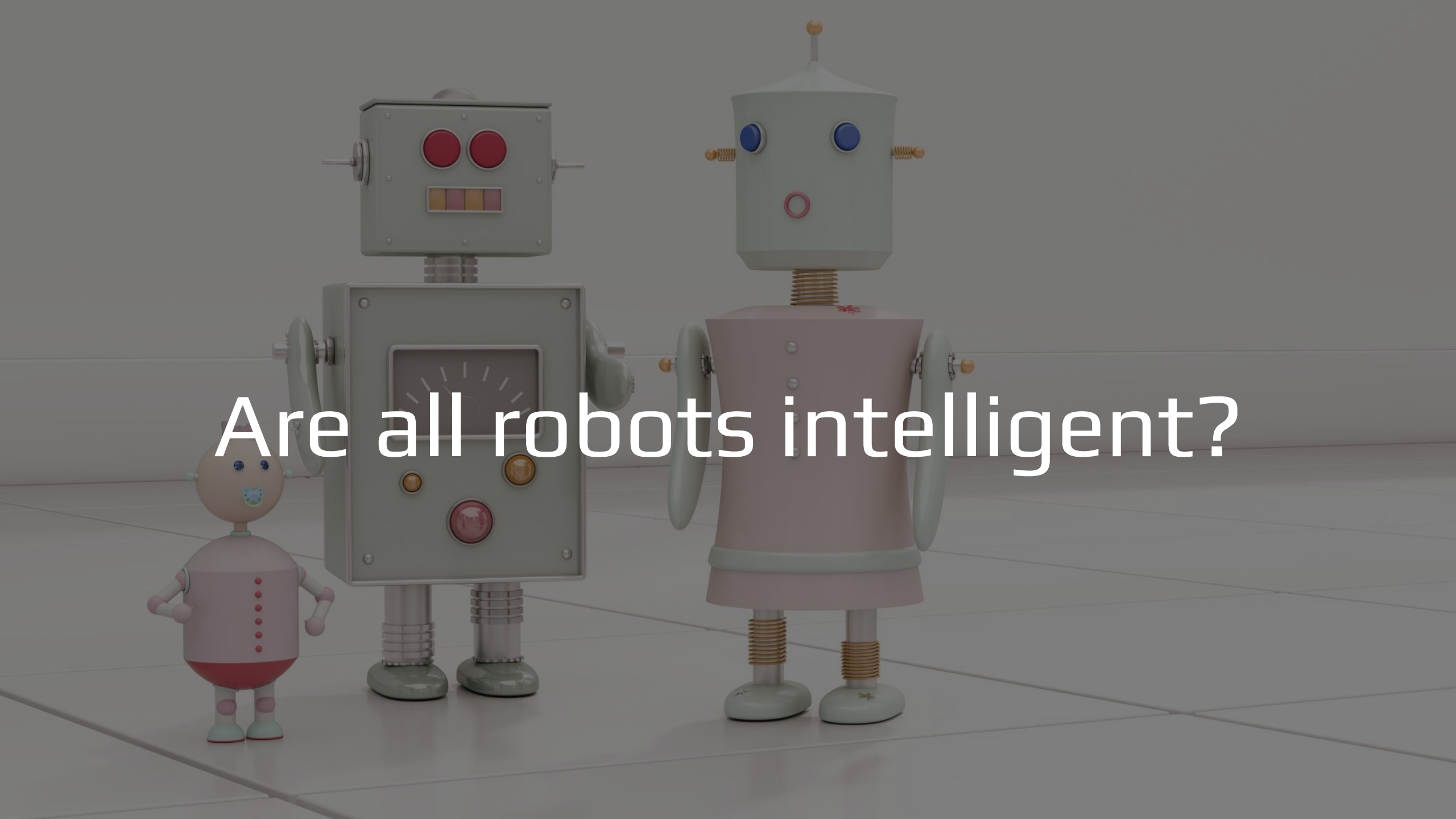
1. **Add Perception (Sensors upgrade)**

- Turbidity sensor (water dirtiness)
- Load/weight sensor (clothes amount)
- Vibration sensor (imbalance detection)
- Vision sensor (optional camera)

2 **Add decision logics (Adaptive control):**

- IF water is still dirty:
  extend wash cycle
- IF imbalance detected:
  redistribute load
- IF clothes are light:
  reduce wash time

3. **Add Feedback Loop:** sense → analyze → adjust → repeat


Are all robots intelligent?

A mobile robot

- Robot has motors
- Robot has distance sensors
- Robot has closed feedback system
- Is it intelligent?

In a nutshell

Automation:

- Do this sequence no matter what
- Microwave timer

Reactive Robot:

- Do this, BUT adjust based on what you sense
- bumper car → reacts instantly to collision

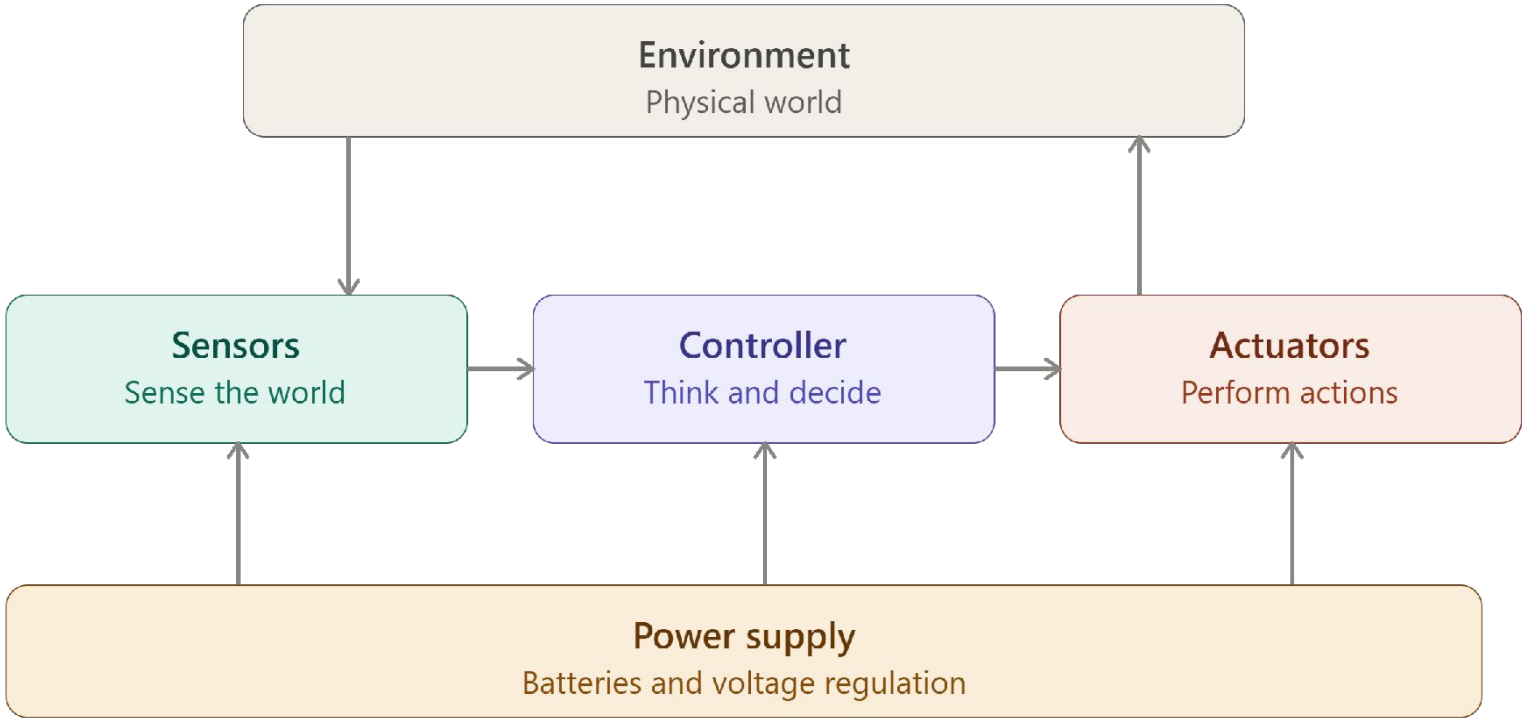
Intelligent Robotics (AI systems)

- Decide what to do based on perception, prediction, and learning
- self-driving car → plans and predicts

Basic Components of a robot

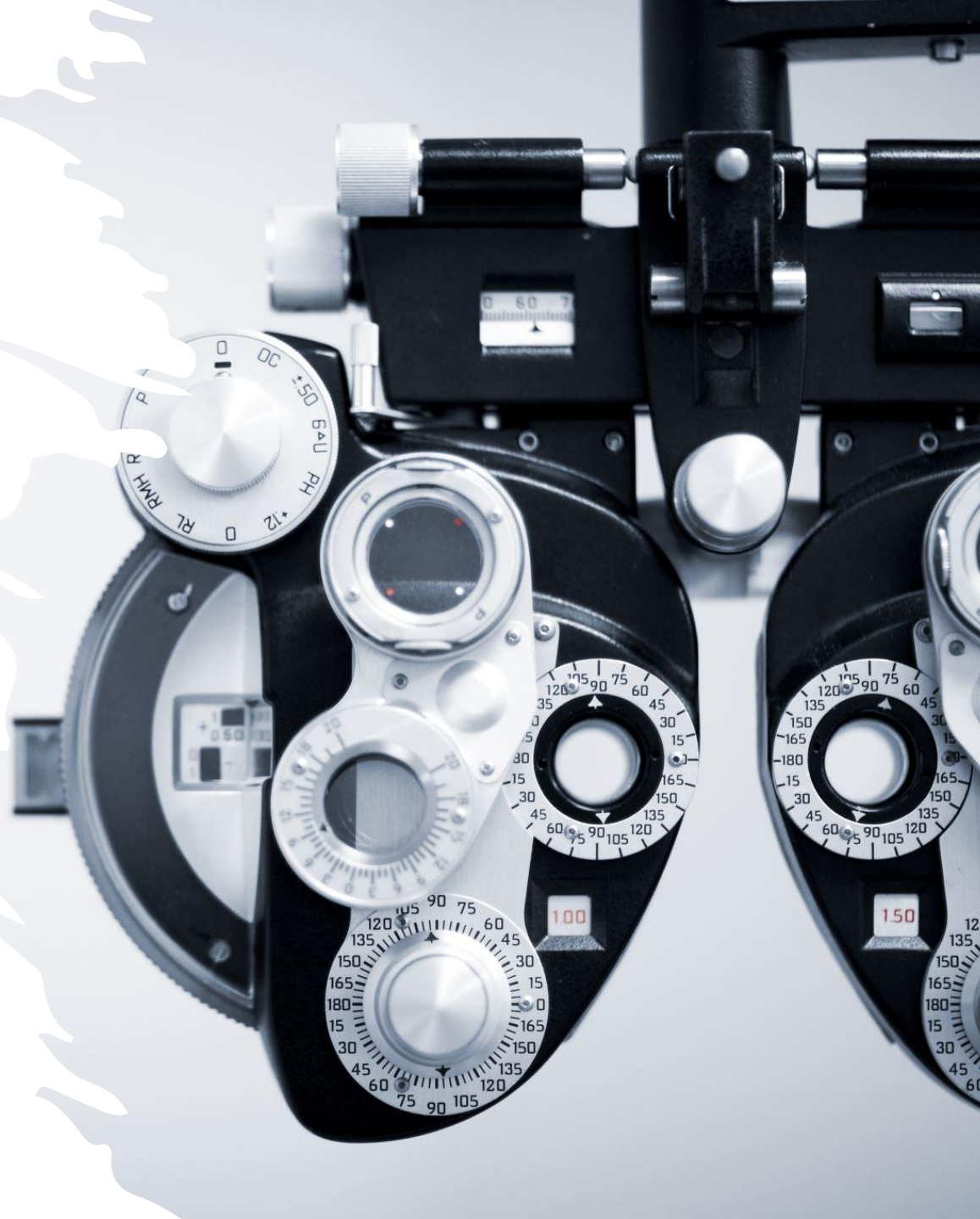
How a Robot Senses, Thinks, and Acts

Basic Robot Architecture



Sensors — *Perception Layer*

- Sensors collect information from the environment.
- **Examples:**
  - Ultrasonic sensor
  - Camera
  - IMU
  - Infrared sensor
  - LiDAR
- **Purpose:**
  - Detect obstacles
  - Measure distance
  - Capture images
  - Monitor orientation and motion
- Sensors act as the eyes and ears of a robot.



Controller — *Decision-Making Layer*

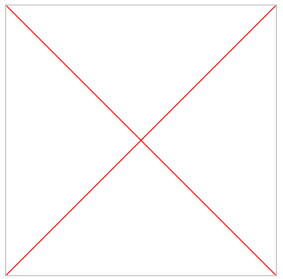
- The controller processes sensor data and decides what actions to perform.
- **Examples:**
  - Arduino
  - ESP32
  - Raspberry Pi
  - Embedded computers
- **Purpose:**
  - Execute control logic
  - Process inputs
  - Run AI or algorithms
  - Coordinate robot behavior

Actuators — *Action Layer*

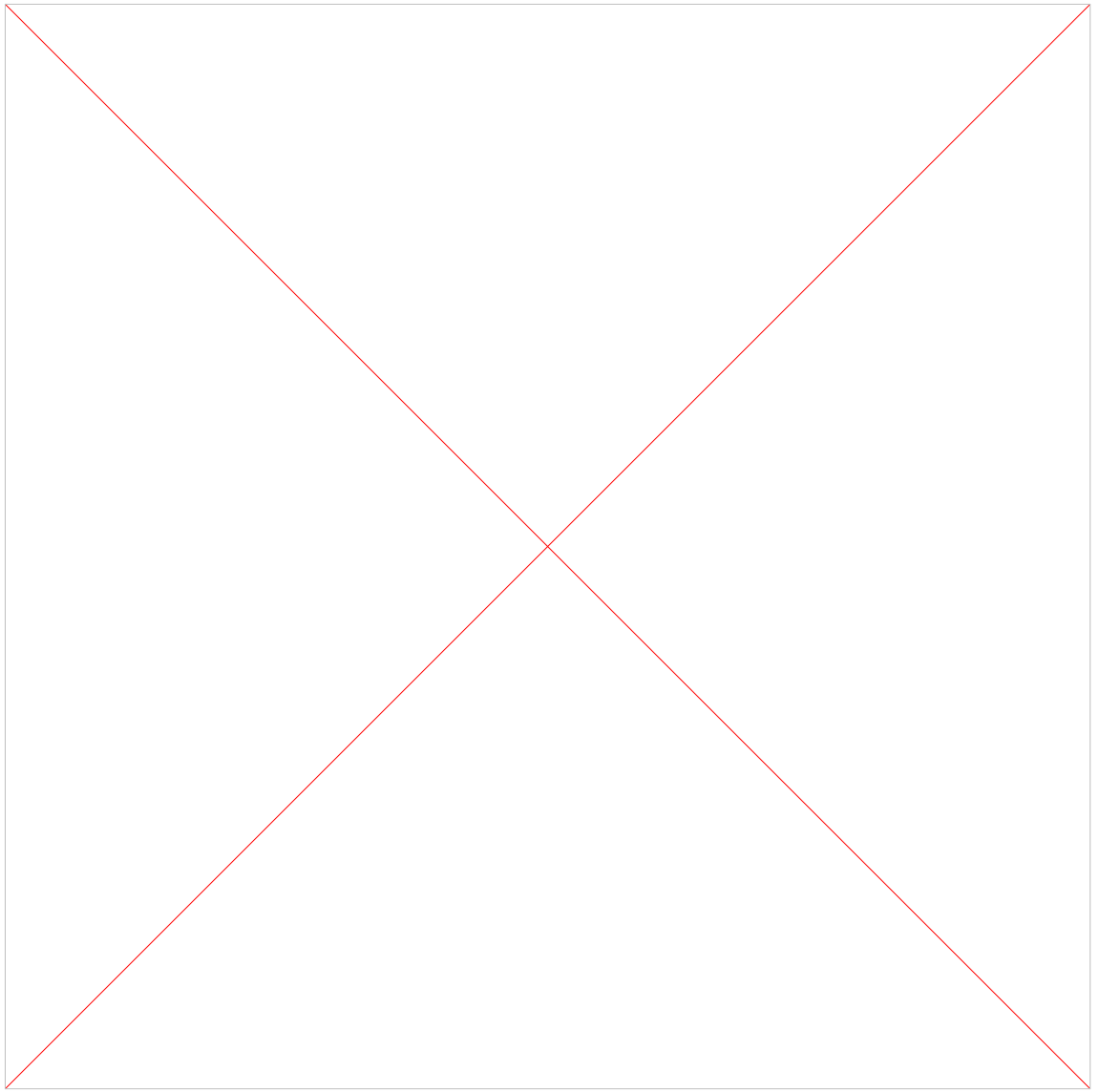
- Actuators convert electrical signals into physical movement.
- **Examples:**
  - DC motors
  - Servo motors
  - Robotic arms
  - Wheels
- **Purpose:**
  - Move the robot
  - Rotate joints
  - Control motion and interaction
- Actuators function as the muscles of a robot.

Basic Robot Workflow

- **Sense → Process → Act**
- Sensors collect data
- Controller analyzes information
- Actuators perform actions



When robotics started?

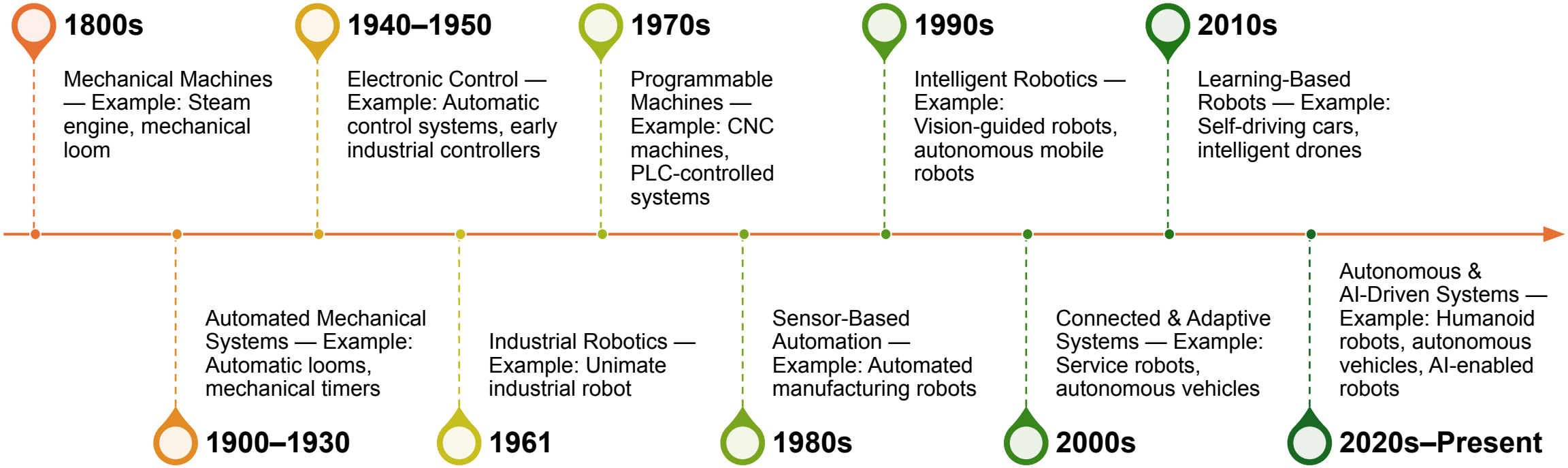


Evolution of Robots

From Mechanical Machines to Intelligent Autonomous Systems



Timeline


1800s
Mechanical Machines — Example: Steam engine, mechanical loom
1900–1930
Automated Mechanical Systems — Example: Automatic looms, mechanical timers
1940–1950
Electronic Control — Example: Automatic control systems, early industrial controllers
1961
Industrial Robotics — Example: Unimate industrial robot
1970s
Programmable Machines — Example: CNC machines, PLC-controlled systems
1980s
Sensor-Based Automation — Example: Automated manufacturing robots
1990s
Intelligent Robotics — Example: Vision-guided robots, autonomous mobile robots
2000s
Connected & Adaptive Systems — Example: Service robots, autonomous vehicles
2010s
Learning-Based Robots — Example: Self-driving cars, intelligent drones
2020s–Present
Autonomous & AI-Driven Systems — Example: Humanoid robots, autonomous vehicles, AI-enabled robots



Ancient Era of Robotics - Mechanical Automatons

- **Key Characteristics**
  - Mechanically operated systems
  - Motion achieved through gears, water, air, or weights
  - Designed mainly for entertainment, religious displays, or scientific curiosity
  - No intelligence or autonomous decision-making
- **Historical Examples**
  - Mechanical birds and moving statues in ancient Greece
  - Water-powered automated devices
  - Early clockwork mechanisms

Industrial Revolution Era - Mechanical Automation Begins

Characteristics:

- Machines powered by steam/electricity
- Repetitive industrial tasks
- No intelligence or sensing

Examples:

- Automated looms
- Mechanical assembly systems

Limitation:

- No perception
- No decision-making
- Fixed operations only

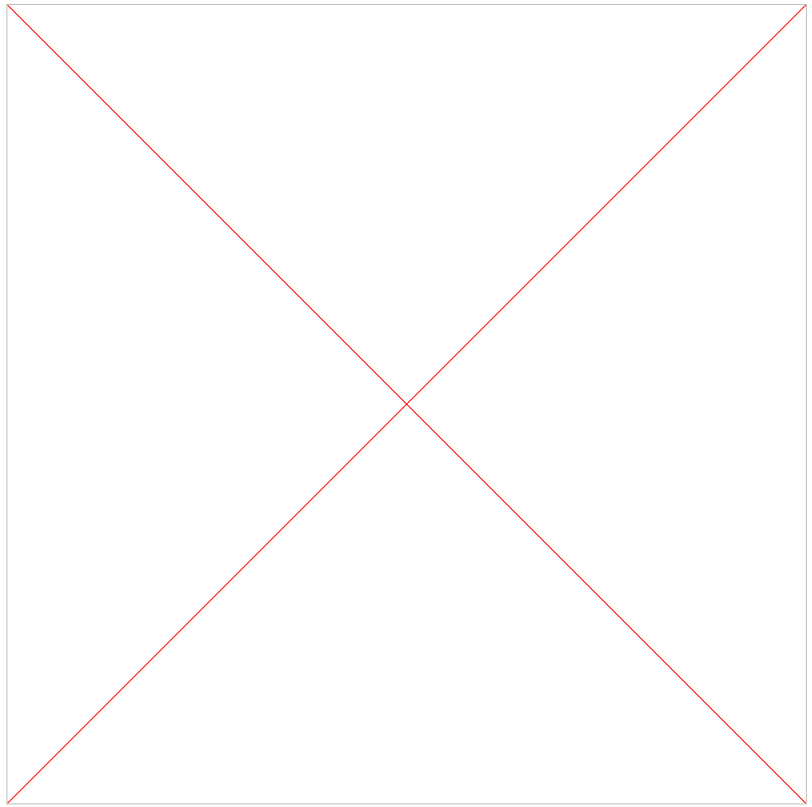
Birth of Industrial Robots (1950s–1970s)

- First Modern Industrial Robots
- **Major Development:** Programmable robotic arms for factories
- **Important Example:** Unimation developed the first industrial robot.
- **Applications:**
  - Welding
  - Pick and place
  - Manufacturing
- **Features:**
  - Programmable
  - Repeatable
  - Limited intelligence



Sensor-Based Robotics (1980s–1990s)

- Robots Begin to Sense the Environment
- **Introduction of:**
  - Ultrasonic sensors
  - Encoders
  - Cameras
  - Infrared sensors
- **Result >** Robots could now:
  - avoid obstacles
  - navigate environments
  - react to surroundings
- **Important Shift:** From fixed automation → reactive robotics



Intelligent Robotics Era (2000s–Present)

Modern robots can:

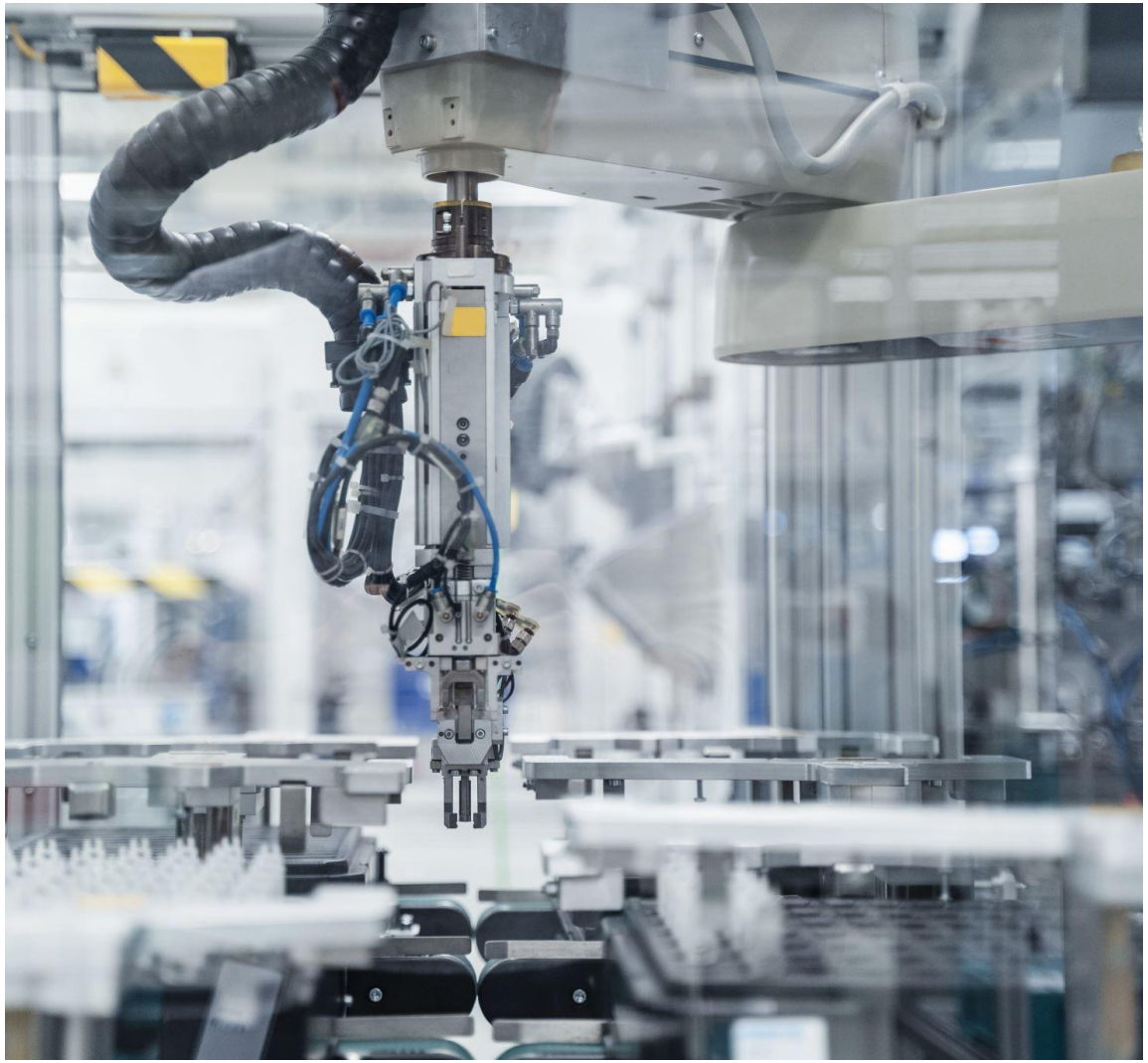
- detect objects
- recognize humans
- navigate autonomously
- make decisions

Technologies:

- Computer Vision
- Machine Learning
- ROS
- SLAM
- Deep Learning

Examples:

- Self-driving cars
- Drones
- Warehouse robots



Modern Robotics Companies

- Boston Dynamics
- Tesla
- NVIDIA
- OpenAI (robotic AI research)

Evolution Timeline

Era	Main Characteristic
Ancient Era	Mechanical automatons
Industrial Era	Fixed automation machines
1950s–70s	Programmable industrial robots
1980s–90s	Sensor-based reactive robots
2000s–Today	AI-powered intelligent robots



Future of Robotics

- Humanoid assistants
- Autonomous transportation
- Medical robotics
- Collaborative robots (cobots)
- AI-driven household robots